MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, NAND GATES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, positive NAND logic gating microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Single, 8-input positive NAND gate
02	Dual, 4-input positive NAND gate
03	Triple, 3-input positive NAND gate
04	Quadruple, 2-input positive NAND gate
05	Hex, 1-input inverter gate
06	Triple, 3-input positive NAND gate (open collector output)
07	Quadruple, 2-input positive NAND gate (open collector output)
08	Hex, 1-input inverter gate (open collector output)
09	Same as device type 07, except different pin connections

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 to +7.0 V
Input voltage range	-1.5 at -12 mA to 5.5 V
Storage temperature range	-65° to +150°C
Maximum power dissipation per gate (P_D) 1/	40 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case A, B, and D	0.09°C/mW
Case C	0.08°C/mW
Junction temperature (T_J)	175°C

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage	4.5 minimum to 5.5 V maximum
Minimum high level input voltage	2.0 V
Maximum low level input voltage	0.8 V
Normalized fanout (each output) 2/	10 maximum
Case operating temperature range	-55° to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagram and terminal connections. Logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Case outlines. The case outlines shall be in accordance with 1.2.3.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

2/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.8 V I_{OH} = -400 µA 1/	01,02, 03,04, 05	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 16 mA V_{IN} = 2.0 V for all inputs of gate under test 1/	All		0.4	Volts
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_C = 25°C	All		-1.5	Volts
Maximum collector cut-off current	I_{CEX}	V_{CC} = 4.5 V, V_{IN} = 0.8 V V_{OH} = 5.5 V	06,07, 08,09,		250	µA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 2/	All		40	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 2/	All		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 1/	All	-0.7	-1.6	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 2/ 3/	01,02, 03,04, 05	-20	-55	mA
High-level supply current per gate	I_{CCH}	V_{CC} = 5.5 V 2/ V_{IN} = 0 V	All		1.65	mA
Low-level supply current per gate	I_{CCL}	V_{CC} = 5.5 V 1/ V_{IN} = 5.5 V	All		5.0	mA
Propagation delay time, high-to-low-level	t_{PHL}	C_L = 50 pF, R_L = 390 Ω	01,02, 03,04, 05	3	24	ns
			06,07, 08,09	3	29	ns
Propagation delay time, low-to-high-level	t_{PLH}	C_L = 50 pF, R_L = 390 Ω	01,02, 03,04, 05	3	27	ns
			06,07, 08,09	3	35	ns

1/ All unspecified inputs at 5.5 volts.
2/ All unspecified inputs grounded.
3/ Not more than one output should be shorted at a time.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 1 (see MIL-M-38510, appendix E).

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3,9, 10,11	1*,2,3,9	1
Group A test requirements (method 5005)	1,2,3, 9,10,11	1,2,3,9	1,2,3,9
Group B test requirements (method 5005, subgroup 5)	1,2,3, 9,10,11	N/A	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3	1
Additional electrical parameters for group C periodic inspections	N/A	10,11	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3	1

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = 125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II; interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection for class B and C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection.

c. Steady state life test (method 1005 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = 125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Inspection of packaging. The inspection of packaging shall be in accordance with MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as specified herein.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

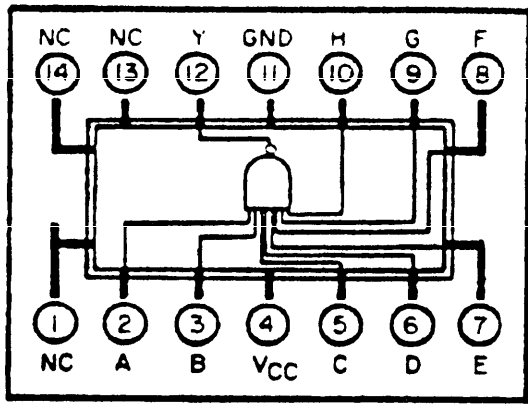
6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

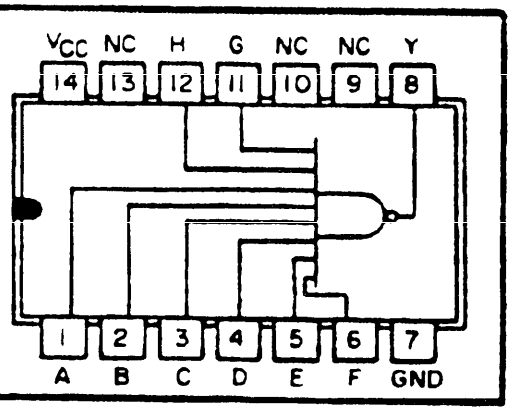
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Device type 01

Case A, B and D

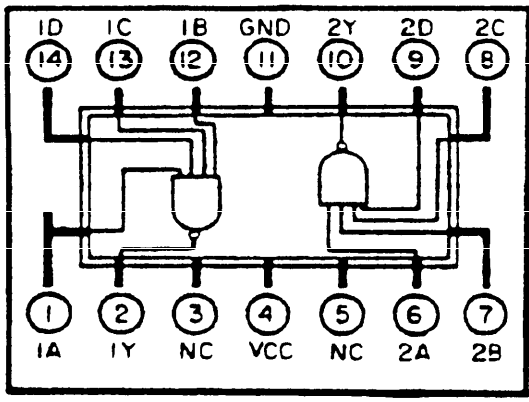


Case C

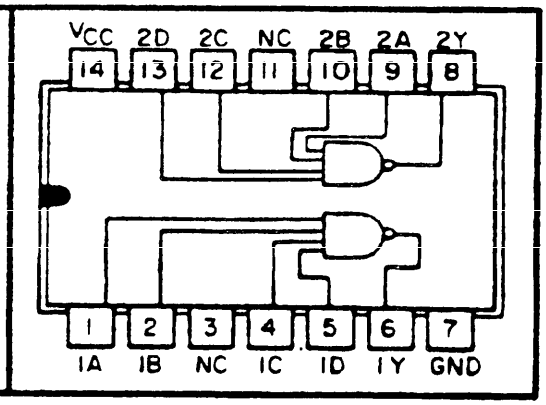


Device type 02

Case A, B and D

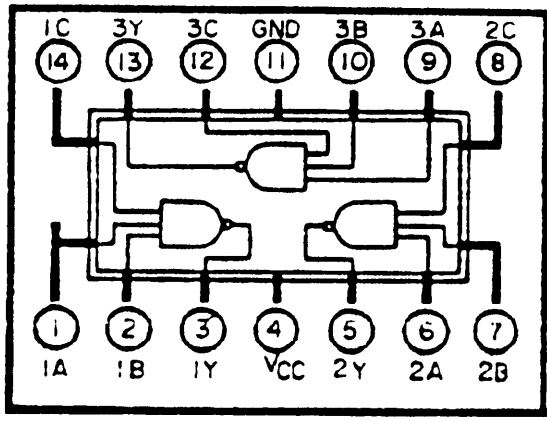


Case C



Device type 03

Case A, B and D



Case C

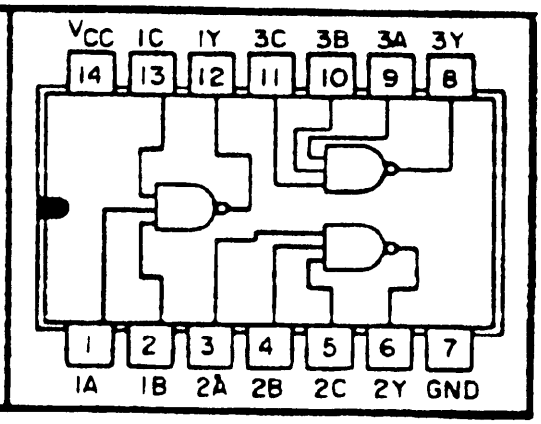
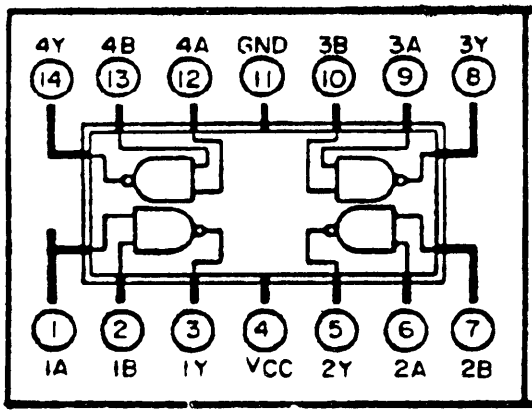


FIGURE 1. Case outlines and terminal connections (top view).

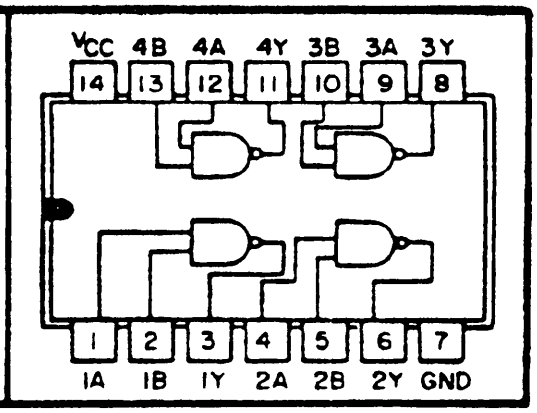
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Device type 04

Case A, B and D

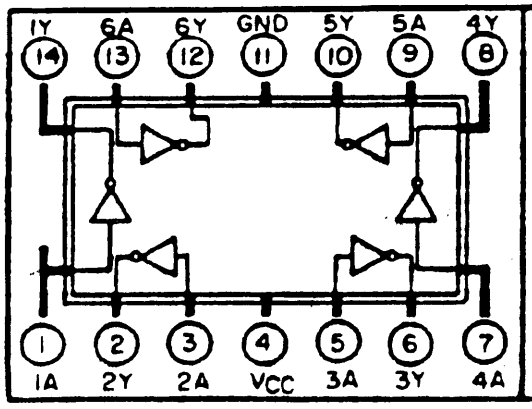


Case C



Device type 05 and 08

Case A, B and D



Case C

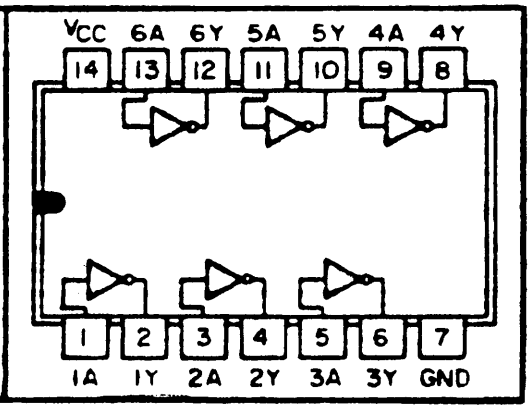
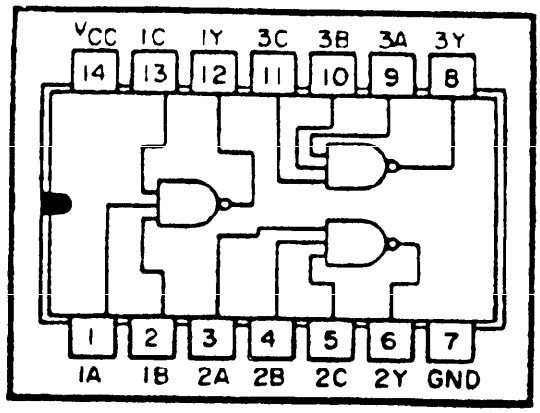


FIGURE 1. Logic diagram and terminal connections (top view) - Continued.

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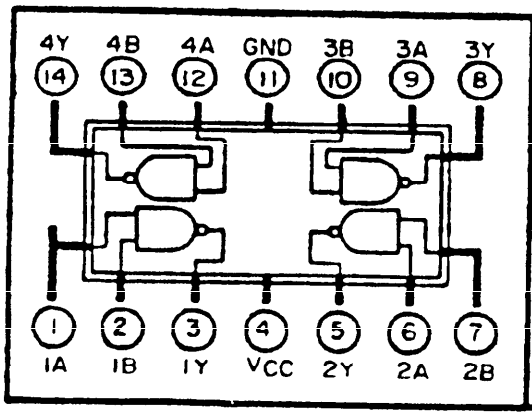
Device type 06

Case A, B, C, and D

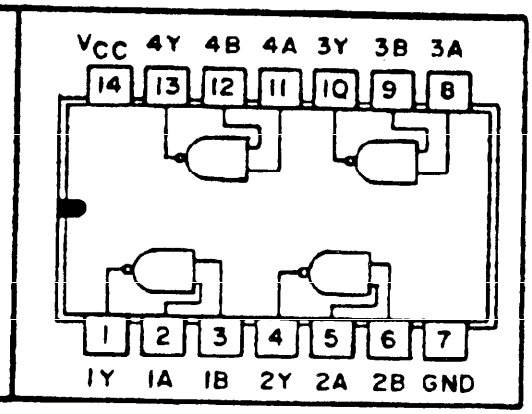


Device type 07

Case A, B and D



Case C



Device type 09

Case C

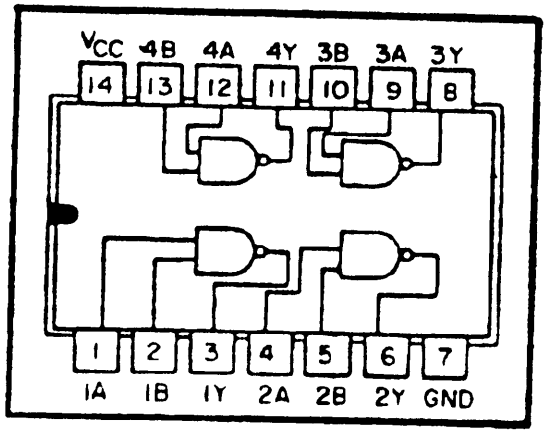


FIGURE 1. Logic diagram and terminal connections (Top Views) - Continued.

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Device Type 01

Truth table								
Input								Output
A	B	C	D	E	F	G	H	Y
H	H	H	H	H	H	H	H	L
All other combinations of H and L at the inputs give H output.								

Positive logic $Y = \overline{ABCDEFGH}$

Device type 03 and 06

Truth table			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positive logic $Y = \overline{ABC}$

Device type 05 and 08

Truth table each gate	
Input	Output
A	Y
L	H
H	L

Positive logic $Y = \overline{A}$

Device type 02

Truth table				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

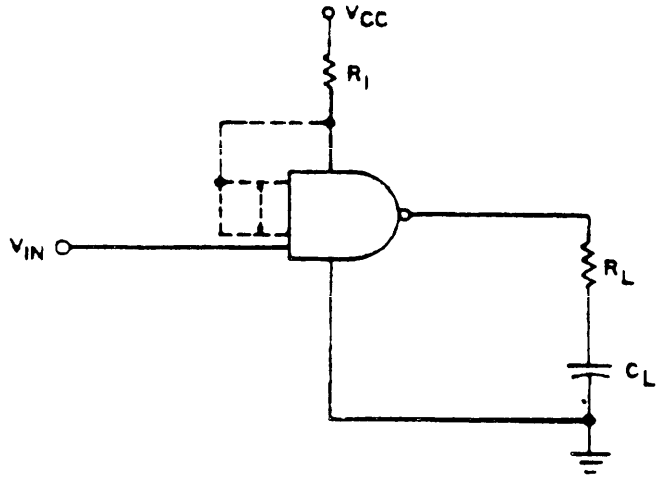
Device type 04, 07 and 09

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

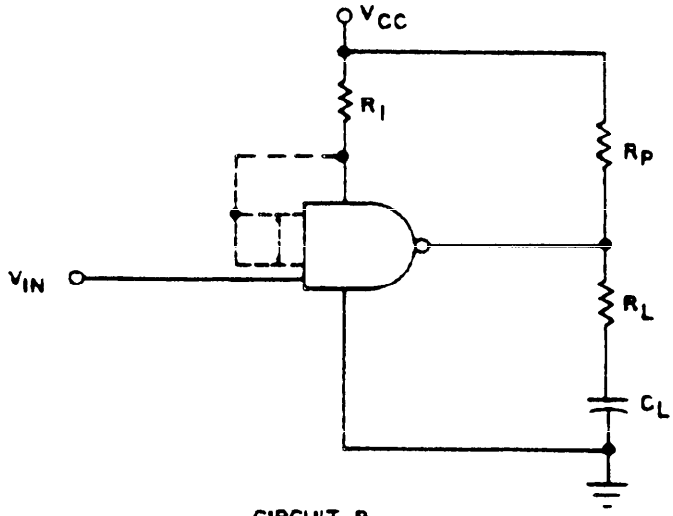
Positive logic $Y = \overline{AB}$

FIGURE 2. Truth tables and logic equations.

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CIRCUIT A



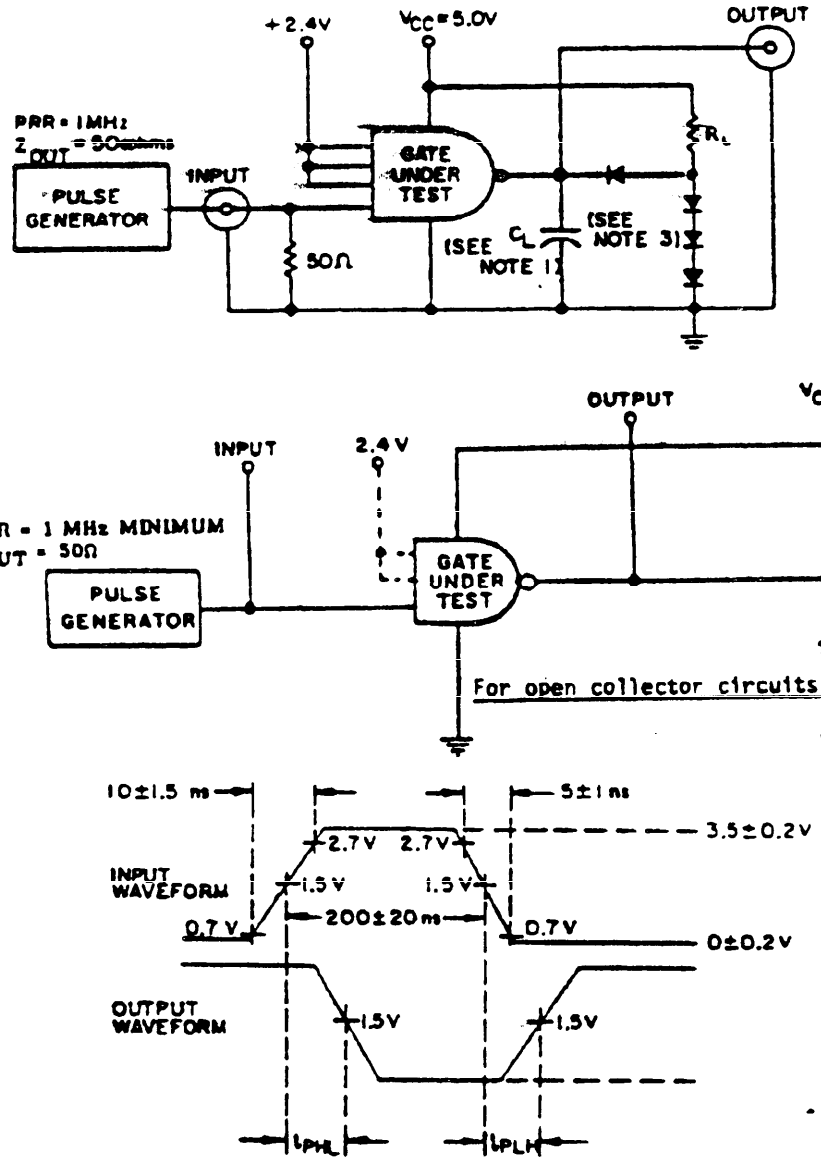
CIRCUIT B

NOTES:
1. One input of each gate shall be driven. The remaining inputs of each gate shall be connected to V_{CC} through the appropriate current limiting resistors.
2. V_{CC} and R_1 are such that the minimum voltage at the device terminals is 5 volts.
3. If parallel excitation is used, the clock frequency shall be 100 KHz with V_{IN} = 3 volts minimum and minimum duty cycle of 50%.
4. R_L = 120Ω ±5%; C_L = .05 µF ±10%; R_p = 300Ω ±5%.
5. Circuit A shall be used for device types 01, 02, 03, 04, and 05. Circuit B shall be used for device types 06, 07, 08, and 09.

FIGURE 3. Burn-in and life test circuits.

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Test circuit except for open collector circuits.



NOTES:
1. C_L = 50 pF minimum, including scope probe, wiring and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 390 ohms ±5 percent.

FIGURE 4. Switching time test circuit and waveforms.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Case C	13	1	2	14	3	4	5	6	11	12	7	8	9	10		Min	Max	
			Test No.	NC	A	B	V_{CC}	C	D	E	F	G	H	GND	Y	NC	NC				
1 T_C= 25°C	V_{OL}	3007	1		2.0 v	2.0 v	4.5 v	2.0 v	2.0 v	2.0 v	2.0 v	2.0 v	2.0 v	GND	16 mA			Y		0.4	V
	V_{OH}	3006	2		0.8 v	5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		-.4 mA			Y	2.4		V
	V_{OH}		3		5.5 v	0.8 v		5.5 v										Y	2.4		V
	V_{OH}		4			5.5 v		0.8 v										Y	2.4		V
	V_{OH}		5					5.5 v	0.8 v									Y	2.4		V
	V_{OH}		6						5.5 v	0.8 v								Y	2.4		V
	V_{OH}		7							5.5 v	0.8 v							Y	2.4		V
	V_{OH}		8								5.5 v	0.8 v						Y	2.4		V
	V_{OH}		9								5.5 v	5.5 v	0.8 v					Y	2.4		V
	I_{OS}	3011	10		GND	GND	5.5 v	GND	GND	GND	GND	GND	GND		GND			Y	-20	-55	mA
	I_{IH1}	3010	11		2.4 v	GND		GND	GND	GND	GND	GND	GND					A		40	μA
	I_{IH1}		12		GND	2.4 v		GND										B		40	μA
	I_{IH1}		13			GND		2.4 v										C		40	μA
	I_{IH1}		14					GND	2.4 v									D		40	μA
	I_{IH1}		15						GND	2.4 v								E		40	μA
	I_{IH1}		16							GND	2.4 v							F		40	μA
	I_{IH1}		17								GND	2.4 v						G		40	μA
	I_{IH1}		18									GND	2.4 v					H		40	μA
	I_{IH2}		19		5.5 v								GND					A		100	μA
	I_{IH2}		20		GND	5.5 v												B		100	μA
	I_{IH2}		21			GND		5.5 v										C		100	μA
	I_{IH2}		22					GND	5.5 v									D		100	μA
	I_{IH2}		23						GND	5.5 v								E		100	μA
	I_{IH2}		24							GND	5.5 v							F		100	μA
	I_{IH2}		25								GND	5.5 v						G		100	μA
	I_{IH2}		26								GND	GND	5.5 v					H		100	μA
	I_{IL}	3009	27		0.4 v	5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v					A	-0.7	-1.6	mA
	I_{IL}		28		5.5 v	0.4 v		5.5 v										B		-1.6	mA
	I_{IL}		29			5.5 v		0.4 v										C		-1.6	mA
	I_{IL}		30					5.5 v	0.4 v									D		-1.6	mA
	I_{IL}		31						5.5 v	0.4 v								E		-1.6	mA
	I_{IL}		32							5.5 v	0.4 v							F		-1.6	mA
	I_{IL}		33								5.5 v	0.4 v						G		-1.6	mA
	I_{IL}		34									5.5 v	0.4 v					H		-1.6	mA

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 13 / NC	2 / 1 / A	3 / 2 / B	4 / 14 / VCC	5 / 3 / C	6 / 4 / D	7 / 5 / E	8 / 6 / F	9 / 11 / G	10 / 12 / H	11 / 7 / GND	12 / 8 / Y	13 / 9 / NC	14 / 10 / NC	Measured terminal	Test limits Min	Test limits Max	Unit
1	I_{CCL}	3005	35		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	GND				V_{CC}		5.0	mA
	I_{CCH}	3005	36		GND	GND	5.5 v	GND	GND	GND	GND	GND	GND					V_{CC}		1.65	mA
	V_{IC}		37		-12 mA		4.5 v											A		-1.5	V
	V_{IC}		38			-12 mA												B		-1.5	V
	V_{IC}		39					-12 mA										C		-1.5	V
	V_{IC}		40						-12 mA									D		-1.5	V
	V_{IC}		41							-12 mA								E		-1.5	V
	V_{IC}		42								-12 mA							F		-1.5	V
	V_{IC}		43									-12 mA						G		-1.5	V
	V_{IC}		44										-12 mA					H		-1.5	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003	45		in	2.4 v	5.0 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	GND	out			A to Y	3	20	ns
T_C=25°C	t_{PLH}	(Fig. 4)	46		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			A to Y	3	25	ns
10	t_{PHL}	3003	47		in	2.4 v	5.0 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	GND	out			A to Y	3	24	ns
T_C=125°C	t_{PLH}	(Fig. 4)	48		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓			A to Y	3	27	ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 6 / 1Y	3 / 3 / NC	4 / 14 / V_{CC}	5 / 11 / NC	6 / 9 / 2A	7 / 10 / 2B	8 / 12 / 2C	9 / 13 / 2D	10 / 8 / 2Y	11 / 7 / GND	12 / 2 / 1B	13 / 4 / 1C	14 / 5 / 1D	Measured terminal	Test limits Min	Test limits Max	Unit
1	V_{OL}	3007	1	2.0 v	16 mA		4.5 v		5.5 v	5.5 v	5.5 v	5.5 v		GND	2.0 v	2.0 v	2.0 v	1Y		0.4	V
T_C = 25°C	V_{OL}	↓	2	5.5 v			↓		2.0 v	2.0 v	2.0 v	2.0 v	16 mA	↓	5.5 v	5.5 v	5.5 v	2Y		0.4	V
	V_{OH}	3006	3	0.8 v	-.4 mA				5.5 v	5.5 v	5.5 v	5.5 v			5.5 v	5.5 v	5.5 v	1Y	2.4		V
	V_{OH}		4	5.5 v	-.4 mA				5.5 v	5.5 v	5.5 v	5.5 v			0.8 v	5.5 v	5.5 v	1Y	2.4		V
	V_{OH}		5	5.5 v	-.4 mA				5.5 v	5.5 v	5.5 v	5.5 v			5.5 v	0.8 v	5.5 v	1Y	2.4		V
	V_{OH}		6	5.5 v	-.4 mA				5.5 v	5.5 v	5.5 v	5.5 v			5.5 v	5.5 v	0.8 v	1Y	2.4		V
	V_{OH}		7	5.5 v					0.8 v	5.5 v	5.5 v	5.5 v	-.4 mA		5.5 v	5.5 v	5.5 v	2Y	2.4		V
	V_{OH}		8	5.5 v					5.5 v	0.8 v	5.5 v	5.5 v	-.4 mA		5.5 v	5.5 v	5.5 v	2Y	2.4		V
	V_{OH}		9	5.5 v					5.5 v	5.5 v	0.8 v	5.5 v	-.4 mA		5.5 v	5.5 v	5.5 v	2Y	2.4		V
	V_{OH}	↓	10	5.5 v			↓		5.5 v	5.5 v	5.5 v	0.8 v	-.4 mA		5.5 v	5.5 v	5.5 v	2Y	2.4		V
	I_{OS}	3011	11	GND	GND		5.5 v								GND	GND	GND	1Y	-20	-55	mA
	I_{OS}	↓	12				↓		GND	GND	GND	GND	GND					2Y	-20	-55	mA
	I_{IH1}	3010	13	2.4 v											GND	GND	GND	1A		40	μA
	I_{IH1}		14	GND											2.4 v	GND	GND	1B		40	μA
	I_{IH1}		15	↓					↓						GND	2.4 v	GND	1C		40	μA
	I_{IH1}		16												↓	GND	2.4 v	1D		40	μA
	I_{IH1}		17						2.4 v							↓	GND	2A		40	μA
	I_{IH1}		18						GND	2.4 v	↓						↓	2B		40	μA
	I_{IH1}		19						↓	GND	2.4 v	↓						2C		40	μA
	I_{IH1}		20	↓					↓	↓	GND	2.4 v			↓	↓	↓	2D		40	μA
	I_{IH2}		21	5.5 v								GND						1A		100	μA
	I_{IH2}		22	GND								↓			5.5 v	↓	↓	1B		100	μA
	I_{IH2}		23	↓					↓						GND	5.5 v	↓	1C		100	μA
	I_{IH2}		24												↓	GND	5.5 v	1D		100	μA
	I_{IH2}		25						5.5 v							↓	GND	2A		100	μA
	I_{IH2}		26						GND	5.5 v	↓						↓	2B		100	μA
	I_{IH2}		27						↓	GND	5.5 v	↓						2C		100	μA
	I_{IH2}	↓	28	↓			↓		↓	↓	GND	5.5 v		↓	↓	↓	↓	2D		100	μA

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1 1 1A	2 6 1Y	3 3 NC	4 14 V_{CC}	5 11 NC	6 9 2A	7 10 2B	8 12 2C	9 13 2D	10 8 2Y	11 7 GND	12 2 1B	13 4 1C	14 5 1D	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	I_{IL}	3009	29	0.4 v			5.5 v		5.5 v	5.5 v	5.5 v	5.5 v		GND	5.5 v	5.5 v	5.5 v	1A	-0.7	-1.6	mA
			30	5.5 v											0.4 v	5.5 v		1B		-1.6	mA
			31												5.5 v	0.4 v		1C		-1.6	mA
			32													5.5 v	0.4 v	1D		-1.6	mA
			33						0.4 v								5.5 v	2A		-1.6	mA
			34						5.5 v	0.4 v								2B		-1.6	mA
			35							5.5 v	0.4 v							2C		-1.6	mA
			36							5.5 v	5.5 v	0.4 v						2D		-1.6	mA
	I_{CCL}	3005	37							5.5 v	5.5 v	5.5 v						V_{CC}		10	mA
	I_{CCH}	3005	38	GND					GND	GND	GND	GND			GND	GND	GND	V_{CC}		3.3	mA
	V_{IC}		39	-12 mA			4.5 v											1A		-1.5	V
	V_{IC}		40												-12 mA			1B		-1.5	V
	V_{IC}		41													-12 mA		1C		-1.5	V
	V_{IC}		42														-12 mA	1D		-1.5	V
	V_{IC}		43						-12 mA									2A		-1.5	V
	V_{IC}		44							-12 mA								2B		-1.5	V
	V_{IC}		45								-12 mA							2C		-1.5	V
	V_{IC}		46									-12 mA						2D		-1.5	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = 25°C	t_{PHL}	3003 (Fig. 4)	47	in	out		5.0 v		in	2.4 v	2.4 v	2.4 v	out	GND	2.4 v	2.4 v	2.4 v	1A to 1Y	3	20	ns
	t_{PHL}		48															2A to 2Y	3	20	ns
	t_{PLH}		49	in	out				in	2.4 v	2.4 v	2.4 v	out		2.4 v	2.4 v	2.4 v	1A to 1Y	3	25	ns
	t_{PLH}		50															2A to 2Y	3	25	ns
10 T_C = 125°C	t_{PHL}	3003 (Fig. 4)	51	in	out		5.0 v		in	2.4 v	2.4 v	2.4 v	out	GND	2.4 v	2.4 v	2.4 v	1A to 1Y	3	24	ns
	t_{PHL}		52															2A to 2Y	3	24	ns
	t_{PLH}		53	in	out				in	2.4 v	2.4 v	2.4 v	out		2.4 v	2.4 v	2.4 v	1A to 1Y	3	27	ns
	t_{PLH}		54															2A to 2Y	3	27	ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 2 / 1B	3 / 12 / 1Y	4 / 14 / V_{CC}	5 / 6 / 2Y	6 / 3 / 2A	7 / 4 / 2B	8 / 5 / 2C	9 / 9 / 3A	10 / 10 / 3B	11 / 7 / GND	12 / 11 / 3C	13 / 8 / 3Y	14 / 13 / 1C	Measured terminal	Test limits Min	Test limits Max	Unit
1	V_{OL}	3007	1	2.0 v	2.0 v	16 mA	4.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	GND	5.5 v		2.0 v	1Y		0.4	V
T_C = 25°C	V_{OL}	↓	2	5.5 v	5.5 v		4.5 v	16 mA	2.0 v	2.0 v	2.0 v	5.5 v	5.5 v	↓	5.5 v		5.5 v	2Y		0.4	V
↓	V_{OL}		3	5.5 v	5.5 v		4.5 v		5.5 v	5.5 v	5.5 v	2.0 v	2.0 v		2.0 v	16 mA	5.5 v	3Y		0.4	V
	V_{OH}	3006	4	0.8 v	5.5 v	-.4 mA	4.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		5.5 v		5.5 v	1Y	2.4		V
	V_{OH}		5	5.5 v	0.8 v	-.4 mA	4.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		5.5 v		5.5 v	1Y	2.4		V
	V_{OH}		6	5.5 v	5.5 v	-.4 mA	4.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		5.5 v		0.8 v	1Y	2.4		V
	V_{OH}		7	↓	↓		↓	-.4 mA	0.8 v	5.5 v	5.5 v	5.5 v	5.5 v		5.5 v		5.5 v	2Y	2.4		V
	V_{OH}		8					-.4 mA	5.5 v	0.8 v	5.5 v	5.5 v	5.5 v		5.5 v		5.5 v	2Y	2.4		V
	V_{OH}		9					-.4 mA	5.5 v	5.5 v	0.8 v	5.5 v	5.5 v		5.5 v		5.5 v	2Y	2.4		V
	V_{OH}		10						5.5 v	5.5 v	5.5 v	0.8 v	5.5 v		5.5 v	-.4 mA	5.5 v	3Y	2.4		V
	V_{OH}		11						5.5 v	5.5 v	5.5 v	5.5 v	0.8 v		5.5 v	-.4 mA	5.5 v	3Y	2.4		V
	V_{OH}	↓	12						5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		0.8 v	-.4 mA	5.5 v	3Y	2.4		V
	I_{OS}	3011	13	GND	GND	GND	5.5 v										GND	1Y	-20	-55	mA
	I_{OS}	↓	14				5.5 v	GND	GND	GND	GND							2Y	-20	-55	mA
	I_{OS}		15				5.5 v					GND	GND		GND	GND		3Y	-20	-55	mA
	I_{IH1}	3010	16	2.4 v	GND		5.5 v		GND	GND	GND	GND	GND		GND		GND	1A		40	μA
	I_{IH1}		17	GND	2.4 v				↓	↓	↓	↓	↓		↓		GND	1B		40	μA
	I_{IH1}		18	↓	GND		↓		2.4 v	↓	↓	↓	↓		↓		2.4 v	1C		40	μA
	I_{IH1}		19		↓				GND	2.4 v	↓	↓	↓		↓		GND	2A		40	μA
	I_{IH1}		20						↓	GND	2.4 v	↓	↓		↓		↓	2B		40	μA
	I_{IH1}		21						↓	↓	GND	2.4 v	↓		↓		↓	2C		40	μA
	I_{IH1}		22						↓	↓	↓	GND	2.4 v		↓		↓	3A		40	μA
	I_{IH1}		23						↓	↓	↓	↓	GND		↓		↓	3B		40	μA
	I_{IH1}		24	↓	↓		↓		↓	↓	↓	↓	GND		2.4 v		↓	3C		40	μA
	I_{IH2}		25	5.5 v	GND		5.5 v											1A		100	μA
	I_{IH2}		26	GND	5.5 v				↓	↓	↓	↓	↓		↓		↓	1B		100	μA
	I_{IH2}		27	↓	GND		↓		5.5 v	↓	↓	↓	↓		↓		5.5 v	1C		100	μA
	I_{IH2}		28		↓				GND	5.5 v	↓	↓	↓		↓		GND	2A		100	μA
	I_{IH2}		29						↓	GND	5.5 v	↓	↓		↓		↓	2B		100	μA
	I_{IH2}		30						↓	↓	GND	5.5 v	↓		↓		↓	2C		100	μA
	I_{IH2}		31						↓	↓	↓	GND	5.5 v		↓		↓	3A		100	μA
	I_{IH2}		32						↓	↓	↓	↓	GND		↓		↓	3B		100	μA
↓	I_{IH2}	↓	33	↓	↓		↓		↓	↓	↓	↓	GND	↓	5.5 v		↓	3C		100	μA

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Case C	1	2	12	14	6	3	4	5	9	10	7	11	8	13				
			Test No.	1A	1B	1Y	V_{CC}	2Y	2A	2B	2C	3A	3B	GND	3C	3Y	1C				
1 T_C = 25°C	I_{IL}	3009	34	0.4 v	5.5 v		5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	GND	5.5 v		5.5 v	1A	-0.7	-1.6	mA
			35	5.5 v	0.4 v												5.5 v	1B			
			36		5.5 v												0.4 v	1C			
			37						0.4 v								5.5 v	2A			
			38						5.5 v	0.4 v								2B			
			39							5.5 v	0.4 v							2C			
			40								5.5 v	0.4 v						3A			
			41									5.5 v	0.4 v					3B			
			42									5.5 v	5.5 v		0.4 v			3C			
	I_{CCH}	3005	43	GND	GND				GND	GND	GND	GND	GND		GND		GND	V_{CC}		4.95	mA
	I_{CCL}	3005	44	5.5 v	5.5 v				5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		5.5 v		5.5 v	V_{CC}		15	mA
	V_{IC}		45	-12 mA			4.5 v											1A		-1.5	V
			46		-12 mA													1B			
			47														-12 mA	1C			
			48						-12 mA									2A			
			49							-12 mA								2B			
			50								-12 mA							2C			
			51									-12 mA						3A			
			52										-12 mA					3B			
			53												-12 mA			3C			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = 25°C	t_{PHL}	3003 (Fig. 4)	54	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	2.4 v	IN	2.4 v	GND	2.4 v	OUT	2.4 v	1A to 1Y	3	20	ns
	t_{PHL}		55															2A to 2Y	3	20	ns
	t_{PHL}		56															3A to 3Y	3	20	ns
	t_{PLH}		57	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y	3	25	ns
	t_{PLH}		58															2A to 2Y	3	25	ns
	t_{PLH}		59															3A to 3Y	3	25	ns
10 T_C = 125°C	t_{PHL}	3003 Fig. 4)	60	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y	3	24	ns
	t_{PHL}		61															2A to 2Y	3	24	ns
	t_{PHL}		62															3A to 3Y	3	24	ns
	t_{PLH}		63	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y	3	27	ns
	t_{PLH}		64															2A to 2Y	3	27	ns
	t_{PLH}		65															3A to 3Y	3	27	ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 04

Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1 1 1A	2 2 1B	3 3 1Y	4 14 V_{CC}	5 6 2Y	6 4 2A	7 5 2B	8 8 3Y	9 9 3A	10 10 3B	11 7 GND	12 12 4A	13 13 4B	14 11 4Y	Measured terminal	Limits Min	Limits Max	Unit
1	V_{OH}	3006	1	2.0 v	2.0 v	16 mA	4.5 v		5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		1Y		0.4	V
T_C = 25°C	V_{OH}		2	5.5 v	5.5 v			16 mA	2.0 v	2.0 v		5.5 v	5.5 v					2Y		0.4	V
	V_{OH}		3						5.5 v	5.5 v	16 mA	2.0 v	2.0 v					3Y		0.4	V
	V_{OH}		4						5.5 v	5.5 v		5.5 v	5.5 v		2.0 v	2.0 v	16 mA	4Y		0.4	V
	V_{OH}	3006	5	0.8 v	5.5 v	-.4 mA			5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1Y	2.4		V
	V_{OH}		6	5.5 v	.8 v	-.4 mA			5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1Y	2.4		V
	V_{OH}		7	5.5 v	5.5 v			-.4 mA	.8 v	5.5 v								2Y	2.4		V
	V_{OH}		8					-.4 mA	5.5 v	.8 v								2Y	2.4		V
	V_{OH}		9							5.5 v	-.4 mA	.8 v	5.5 v					3Y	2.4		V
	V_{OH}		10								-.4 mA	5.5 v	.8 v					3Y	2.4		V
	V_{OH}		11										5.5 v		.8 v	5.5 v	-.4 mA	4Y	2.4		V
	V_{OH}		12										5.5 v		5.5 v	.8 v	-.4 mA	4Y	2.4		V
	I_{OS}	3011	13	GND	GND	GND	5.5 v											1Y	-20	-55	mA
	I_{OS}		14					GND	GND	GND								2Y	-20	-55	mA
	I_{OS}		15								GND	GND	GND					3Y	-20	-55	mA
	I_{OS}		16												GND	GND	GND	4Y	-20	-55	mA
	I_{IH1}	3010	17	2.4 v	GND				GND	GND		GND	GND		GND	GND		1A		40	µA
	I_{IH1}		18	GND	2.4 v				GND									1B		40	µA
	I_{IH1}		19		GND				2.4 v									2A		40	µA
	I_{IH1}		20						GND	2.4 v								2B		40	µA
	I_{IH1}		21							GND		2.4 v						3A		40	µA
	I_{IH1}		22									GND	2.4 v					3B		40	µA
	I_{IH1}		23										GND		2.4 v			4A		40	µA
	I_{IH1}		24												GND	2.4 v		4B		40	µA
	I_{IH2}		25	5.5 v												GND		1A		100	µA
	I_{IH2}		26	GND	5.5 v													1B		100	µA
	I_{IH2}		27		GND				5.5 v									2A		100	µA
	I_{IH2}		28						GND	5.5 v								2B		100	µA
	I_{IH2}		29							GND		5.5 v						3A		100	µA
	I_{IH2}		30									GND	5.5 v					3B		100	µA
	I_{IH2}		31										GND		5.5 v			4A		100	µA
	I_{IH2}		32												GND	5.5 v		4B		100	µA
	I_{IL}	3009	33	0.4 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1A	-0.7	-1.6	mA
	I_{IL}		34	5.5 v	0.4 v				5.5 v	5.5 v								1B		-1.6	mA
	I_{IL}		35		5.5 v				0.4 v	5.5 v								2A		-1.6	mA
	I_{IL}		36						5.5 v	0.4 v								2B		-1.6	mA
	I_{IL}		37							5.5 v		0.4 v	5.5 v					3A		-1.6	mA
	I_{IL}		38									5.5 v	0.4 v					3B		-1.6	mA
	I_{IL}		39										5.5 v		0.4 v			4A		-1.6	mA
	I_{IL}		40												5.5 v	0.4 v		4B		-1.6	mA

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated are open)

Sub-group	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 4 / 2Y	3 / 3 / 2A	4 / 14 / V_{CC}	5 / 5 / 3A	6 / 6 / 3Y	7 / 9 / 4A	8 / 8 / 4Y	9 / 11 / 5A	10 / 10 / 5Y	11 / 7 / GND	12 / 12 / 6Y	13 / 13 / 6A	14 / 2 / 1Y	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C 25°C	V_{OL}	3007	1	2.0 v		5.5 v	4.5 v	5.5 v		5.5 v		5.5 v		GND		5.5 v	16 mA	1Y		0.4	V
	V_{OL}		2	5.5 v	16 mA	2.0 v		5.5 v										2Y		0.4	V
	V_{OL}		3			5.5 v		2.0 v	16 mA	2.0 v	16 mA							3Y		0.4	V
	V_{OL}		4					5.5 v		5.5 v		2.0 v	16 mA					4Y		0.4	V
	V_{OL}		5									5.5 v						5Y		0.4	V
	V_{OL}		6												16 mA	2.0 v		6Y		0.4	V
	V_{OH}	3006	7	0.8 v		5.5 v										5.5 v	-.4 mA	1Y	2.4		V
	V_{OH}		8	5.5 v	-.4 mA	0.8 v		0.8 v	-.4 mA									2Y	2.4		V
	V_{OH}		9			5.5 v		5.5 v		0.8 v	-.4 mA							3Y	2.4		V
	V_{OH}		10							5.5 v		0.8 v	-.4 mA					4Y	2.4		V
	V_{OH}		11							5.5 v		5.5 v						5Y	2.4		V
	V_{OH}		12												-.4 mA	0.8 v		6Y	2.4		V
	I_{OS}	3011	13	GND			5.5 v										GND	1Y	-20	-55	mA
	I_{OS}		14		GND	GND		GND	GND									2Y	-20	-55	mA
	I_{OS}		15							GND	GND							3Y	-20	-55	mA
	I_{OS}		16									GND	GND					4Y	-20	-55	mA
	I_{OS}		17												GND	GND		5Y	-20	-55	mA
	I_{OS}		18															6Y	-20	-55	mA
	I_{IH1}	3010	19	2.4 v		GND		GND		GND		GND						1A		40	µA
	I_{IH1}		20	GND		2.4 v		GND										2A		40	µA
	I_{IH1}		21			GND		2.4 v		2.4 v								3A		40	µA
	I_{IH1}		22					GND		GND		2.4 v						4A		40	µA
	I_{IH1}		23									GND				2.4 v		5A		40	µA
	I_{IH1}		24															6A		40	µA
	I_{IH2}		25	5.5 v												GND		1A		100	µA
	I_{IH2}		26	GND		5.5 v		5.5 v										2A		100	µA
	I_{IH2}		27			GND		GND		5.5 v								3A		100	µA
	I_{IH2}		28							GND		5.5 v						4A		100	µA
	I_{IH2}		29							GND		GND				5.5 v		5A		100	µA
	I_{IH2}		30															6A		100	µA
	I_{IL}	3009	31	0.4 v		5.5 v		5.5 v		5.5 v		5.5 v						1A	-0.7	-1.6	mA
	I_{IL}		32	5.5 v		0.4 v		5.5 v										2A		-1.6	mA
	I_{IL}		33			5.5 v		0.4 v		0.4 v								3A		-1.6	mA
	I_{IL}		34					5.5 v		5.5 v		0.4 v						4A		-1.6	mA
	I_{IL}		35							5.5 v		5.5 v				0.4 v		5A		-1.6	mA
	I_{IL}		36															6A		-1.6	mA

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 / 1 / 1A	2 / 4 / 2Y	3 / 3 / 2A	4 / 14 / V_{CC}	5 / 5 / 3A	6 / 6 / 3Y	7 / 9 / 4A	8 / 8 / 4Y	9 / 11 / 5A	10 / 10 / 5Y	11 / 7 / GND	12 / 12 / 6Y	13 / 13 / 6A	14 / 2 / 1Y	Measured terminal	Test limits Min	Test limits Max	Unit
1 $T_C = 25°C$	V_{OL}	3007	1	2.0 v		5.5 v	4.5 v	5.5 v		5.5 v		5.5 v		GND		5.5 v	16 mA	1Y		0.4	V
	V_{OL}		2	5.5 v	16 mA	2.0 v		5.5 v										2Y		0.4	V
	V_{OL}		3			5.5 v		2.0 v	16 mA									3Y		0.4	V
	V_{OL}		4					5.5 v		2.0 v	16 mA							4Y		0.4	V
	V_{OL}		5							5.5 v		2.0 v	16 mA					5Y		0.4	V
	V_{OL}		6									5.5 v			16 mA	2.0 v		6Y		0.4	V
	V_{OH}	3006	7	0.8 v		5.5 v										5.5 v	-.4 mA	1Y	2.4		V
	V_{OH}		8	5.5 v	-.4 mA	0.8 v		0.8 v	-.4 mA									2Y	2.4		V
	V_{OH}		9			5.5 v		5.5 v		0.8 v	-.4 mA							3Y	2.4		V
	V_{OH}		10							5.5 v		0.8 v	-.4 mA					4Y	2.4		V
	V_{OH}		11							5.5 v		5.5 v			-.4 mA	0.8 v		5Y	2.4		V
	V_{OH}		12															6Y	2.4		V
	I_{OS}	3011	13	GND			5.5 v										GND	1Y	-20	-55	mA
	I_{OS}		14		GND	GND												2Y	-20	-55	mA
	I_{OS}		15					GND	GND									3Y	-20	-55	mA
	I_{OS}		16							GND	GND							4Y	-20	-55	mA
	I_{OS}		17									GND	GND					5Y	-20	-55	mA
	I_{OS}		18												GND	GND		6Y	-20	-55	mA
	I_{IH1}	3010	19	2.4 v		GND		GND		GND		GND						1A		40	µA
	I_{IH1}		20	GND		2.4 v		GND										2A		40	µA
	I_{IH1}		21			GND		2.4 v										3A		40	µA
	I_{IH1}		22					GND		2.4 v								4A		40	µA
	I_{IH1}		23							GND		2.4 v						5A		40	µA
	I_{IH1}		24									GND				2.4 v		6A		40	µA
	I_{IH2}		25	5.5 v												GND		1A		100	µA
	I_{IH2}		26	GND		5.5 v												2A		100	µA
	I_{IH2}		27			GND		5.5 v										3A		100	µA
	I_{IH2}		28					GND		5.5 v								4A		100	µA
	I_{IH2}		29							GND		5.5 v						5A		100	µA
	I_{IH2}		30							GND		GND				5.5 v		6A		100	µA
	I_{IL}	3009	31	0.4 v		5.5 v		5.5 v		5.5 v		5.5 v						1A	-0.7	-1.6	mA
	I_{IL}		32	5.5 v		0.4 v		5.5 v										2A		-1.6	mA
	I_{IL}		33			5.5 v		0.4 v										3A		-1.6	mA
	I_{IL}		34					5.5 v		0.4 v								4A		-1.6	mA
	I_{IL}		35							5.5 v		0.4 v						5A		-1.6	mA
	I_{IL}		36							5.5 v		5.5 v				0.4 v		6A		-1.6	mA

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 4 / 2Y	3 / 3 / 2A	4 / 14 / V_{CC}	5 / 5 / 3A	6 / 6 / 3Y	7 / 9 / 4A	8 / 8 / 4Y	9 / 11 / 5A	10 / 10 / 5Y	11 / 7 / GND	12 / 12 / 6Y	13 / 13 / 6A	14 / 2 / 1Y	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	I_{CCL}	3005	37	5.5 v		5.5 v	5.5 v	5.5 v		5.5 v		5.5 v		GND		5.5 v		V_{CC}		30	mA
	I_{CCH}	3005	38	GND		GND	5.5 v	GND		GND		GND				GND		V_{CC}		9.9	mA
	V_{IC}		39	-12 mA			4.5 v											1A		-1.5	V
			40			-12 mA												2A			
			41					-12 mA										3A			
			42							-12 mA								4A			
			43									-12 mA						5A			
			44													-12 mA		6A			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = 25°C	t_{PHL}	3003 (Fig. 4)	45	IN			5.0 v							GND			OUT	1A to 1Y	3	20	ns
			46		OUT	IN		IN	OUT									2A to 2Y			
			47							IN	OUT							3A to 3Y			
			48									IN	OUT					4A to 4Y			
			49												OUT	IN		5A to 5Y			
			50															6A to 6Y			
	t_{PLH}		51	IN													OUT	1A to 1Y	3	25	ns
			52		OUT	IN		IN	OUT									2A to 2Y			
			53							IN	OUT							3A to 3Y			
			54									IN	OUT					4A to 4Y			
			55												OUT	IN		5A to 5Y			
			56															6A to 6Y			
10 T_C = 125°C	t_{PHL}	3003 (Fig. 4)	57	IN			5.0 v							GND			OUT	1A to 1Y	3	24	ns
	t_{PHL}		58		OUT	IN		IN	OUT									2A to 2Y	3	24	ns
	t_{PHL}		59							IN	OUT							3A to 3Y	3	24	ns
	t_{PHL}		60									IN	OUT					4A to 4Y	3	24	ns
	t_{PHL}		61												OUT	IN		5A to 5Y	3	24	ns
	t_{PHL}		62															6A to 6Y	3	24	ns
	t_{PLH}		63	IN													OUT	1A to 1Y	3	27	ns
	t_{PLH}		64		OUT	IN		IN	OUT									2A to 2Y	3	27	ns
	t_{PLH}		65							IN	OUT							3A to 3Y	3	27	ns
	t_{PLH}		66									IN	OUT					4A to 4Y	3	27	ns
	t_{PLH}		67												OUT	IN		5A to 5Y	3	27	ns
	t_{PLH}		68															6A to 6Y	3	27	ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C, and D	1	2	12	14	6	3	4	5	9	10	7	11	8	13	Measured terminal	Test limits Min	Test limits Max	Unit
			Test No.	1A	1B	1Y	V_{CC}	2Y	2A	2B	2C	3A	3B	GND	3C	3Y	1C				
T_C = 25°C	V_{OL}	3007	1	2.0 V	2.0 V	16 mA	4.5 V	16 mA	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V	5.5 V	2.0 V	1Y		0.4	V
	V_{OL}	3007	2	5.5 V	5.5 V				2.0 V	2.0 V	2.0 V	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	2Y		0.4	V
	V_{OL}	3007	3	5.5 V	5.5 V				5.5 V	5.5 V	5.5 V	2.0 V	2.0 V		2.0 V	16 mA		3Y		0.4	V
	I_{CEX}		4	0.8 V	5.5 V	5.5 V			5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V		5.5 V	1Y		250	µA
			5	5.5 V	0.8 V	5.5 V			2.0 V	2.0 V	2.0 V	5.5 V	5.5 V		5.5 V		5.5 V	1Y		250	µA
			6		5.5 V	5.5 V			5.5 V	5.5 V	5.5 V	2.0 V	2.0 V		2.0 V		0.8 V	1Y		250	µA
			7					5.5 V	0.8 V	0.8 V	0.8 V	5.5 V	5.5 V		5.5 V		5.5 V	2Y		250	µA
			8					5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V			2Y		250	µA
			9					5.5 V			0.8 V	0.8 V	0.8 V			5.5 V		2Y		250	µA
			10								5.5 V	5.5 V	5.5 V			5.5 V		3Y		250	µA
			11									0.8 V	0.8 V			5.5 V		3Y		250	µA
			12										5.5 V			5.5 V		3Y		250	µA
	V_{IC}		13	-12 mA	-12 mA												-12 mA	1A		-1.5	V
			14															1B		-1.5	V
			15						-12 mA									1C		-1.5	V
			16							-12 mA								2A		-1.5	V
			17								-12 mA							2B		-1.5	V
			18									-12 mA						2C		-1.5	V
			19										-12 mA					3A		-1.5	V
			20												-12mA			3B		-1.5	V
			21															3C		-1.5	V
	I_{IH1}	3010	22	2.4 V	GND		5.5 V		GND	GND	GND	GND	GND		GND		GND	1A		40	µA
			23	GND	2.4 V				GND	GND	GND	GND	GND				GND	1B		40	µA
			24		GND				GND	GND	GND	GND	GND				2.4 V	1C		40	µA
			25						2.4 V	GND	2.4 V	2.4 V	2.4 V				GND	2A		40	µA
			26						GND	GND	GND	GND	GND					2B		40	µA
			27							GND	GND	GND	GND					2C		40	µA
			28									GND	GND					3A		40	µA
			29										GND					3B		40	µA
			30															3C		40	µA
	I_{IH2}	3010	31	5.5 V	GND				GND	GND	GND	GND	GND		GND		GND	1A		100	µA
			32	GND	5.5 V				GND	GND	GND	GND	GND				GND	1B		100	µA
			33		GND				GND	GND	GND	GND	GND				5.5 V	1C		100	µA
			34						5.5 V	GND	5.5 V	5.5 V	5.5 V				GND	2A		100	µA
			35						GND	GND	GND	GND	GND				GND	2B		100	µA
			36						GND	GND		GND	GND				GND	2C		100	µA

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C, and D	1	2	12	14	6	3	4	5	9	10	7	11	8	13	Measured terminal	Test limits Min	Test limits Max	Unit
			Test No.	1A	1B	1Y	V_{CC}	2Y	2A	2B	2C	3A	3B	GND	3C	3Y	1C				
1 T_C = 25°C	I_{IH2}	3010	37	GND	GND		5.5 v		GND	GND	GND	5.5 v	GND	GND	GND		GND	3A		100	µA
	I_{IH2}	3010	38	GND	GND				GND	GND	GND	GND	5.5 v		GND		GND	3B		100	µA
	I_{IH2}	3010	39	GND	GND				GND	GND	GND	GND	GND		5.5 v		GND	3C		100	µA
	I_{IL}	3009	40	0.4 v	5.5 v				5.5 v	5.5 v	5.5 v	5.5 v					5.5 v	1A	-0.7	-1.6	mA
			41	5.5 v	0.4 v				5.5 v								5.5 v	1B		-1.6	mA
			42		5.5 v				5.5 v								0.4 v	1C		-1.6	mA
			43						0.4 v								5.5 v	2A		-1.6	mA
			44						5.5 v	0.4 v								2B		-1.6	mA
			45							5.5 v	0.4 v							2C		-1.6	mA
			46								5.5 v	0.4 v						3A		-1.6	mA
			47									5.5 v	0.4 v					3B		-1.6	mA
			48										5.5 v		0.4 v			3C		-1.6	mA
	I_{CCL}	3005	49										5.5 v		5.5 v			V_{CC}		15	mA
	I_{CCH}	3005	50	GND	GND				GND	GND	GND	GND	GND		GND		GND	V_{CC}		4.95	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = 25°C	t_{PHL}	3003 (Fig 4)	51	IN	2.4 v	OUT	5.0 v							GND			2.4 v	1A to 1Y	3	23	ns
	t_{PHL}		52					OUT	IN	2.4 v	2.4 v							2A to 2Y		23	ns
	t_{PHL}		53									IN	2.4 v		2.4 v	OUT		3A to 3Y		23	ns
	t_{PLH}		54	IN	2.4 v	OUT											2.4 v	1A to 1Y		28	ns
	t_{PLH}		55					OUT	IN	2.4 v	2.4 v							2A to 2Y		28	ns
	t_{PLH}		56									IN	2.4 v		2.4 v	OUT		3A to 3Y		28	ns
10 T_C = 125°C	t_{PHL}	3003 (Fig 4)	57	IN	2.4 v	OUT											2.4 v	1A to 1Y		29	ns
	t_{PHL}		58					OUT	IN	2.4 v	2.4 v							2A to 2Y		29	ns
	t_{PHL}		59									IN	2.4 v		2.4 v	OUT		3A to 3Y		29	ns
	t_{PLH}		60	IN	2.4 v	OUT											2.4 v	1A to 1Y		35	ns
	t_{PLH}		61					OUT	IN	2.4 v	2.4 v							2A to 2Y		35	ns
	t_{PLH}		62									IN	2.4 v		2.4 v	OUT		3A to 3Y		35	ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 07.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Case C	2	3	1	14	4	5	6	10	8	9	7	11	12	13				
			Test No.	1A	1B	1Y	V_{CC}	2Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y				
1 T_C = 25°C	V_{OL}	3007	1	2.0 v	2.0 v	16 mA	4.5 v		5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		1Y		0.4	V
			2	5.5 v	5.5 v			16 mA	2.0 v	2.0 v		5.5 v	5.5 v		5.5 v	5.5 v		2Y		0.4	V
			3	5.5 v	5.5 v				5.5 v	5.5 v	16 mA	2.0 v	2.0 v		5.5 v	5.5 v		3Y		0.4	V
			4	5.5 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		2.0 v	2.0 v	16 mA	4Y		0.4	V
	I_{CEX}		5	0.8 v	4.5 v	5.5 v			5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1Y		250	µA
			6	4.5 v	0.8 v	5.5 v			5.5 v	5.5 v					5.5 v	5.5 v		1Y		250	µA
			7	5.5 v	5.5 v			5.5 v	0.8 v	4.5 v								2Y		250	µA
			8					5.5 v	4.5 v	0.8 v								2Y		250	µA
			9						5.5 v	5.5 v	5.5 v	0.8 v	4.5 v					3Y		250	µA
			10								5.5 v	4.5 v	0.8 v					3Y		250	µA
			11									5.5 v	5.5 v		0.8 v	4.5 v	5.5 v	4Y		250	µA
			12									5.5 v	5.5 v		4.5 v	0.8 v	5.5 v	4Y		250	µA
	I_{IH1}	3010	13	2.4 v	GND		5.5 v		GND	GND		GND	GND		GND	GND		1A		40	µA
			14	GND	2.4 v				GND	GND								1B		40	µA
			15		GND				2.4 v	GND								2A		40	µA
			16						GND	2.4 v								2B		40	µA
			17							GND		2.4 v						3A		40	µA
			18									GND	2.4 v					3B		40	µA
			19										GND		2.4 v			4A		40	µA
			20												GND	2.4 v		4B		40	µA
	I_{IH2}	3010	21	5.5 v	GND				GND	GND		GND	GND		GND	GND		1A		100	µA
			22	GND	5.5 v				GND	GND								1B		100	µA
			23		GND				5.5 v	GND								2A		100	µA
			24						GND	5.5 v								2B		100	µA
			25							GND		5.5 v						3A		100	µA
			26									GND	5.5 v					3B		100	µA
			27										GND		5.5 v			4A		100	µA
			28												GND	5.5 v		4B		100	µA
	I_{IL}	3009	29	0.4 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1A	-0.7	-1.6	mA
			30	5.5 v	0.4 v				5.5 v									1B		-1.6	mA
			31		5.5 v				0.4 v									2A		-1.6	mA
			32						5.5 v	0.4 v								2B		-1.6	mA
			33							5.5 v		0.4 v						3A		-1.6	mA
			34									5.5 v	0.4 v					3B		-1.6	mA
			35										5.5 v		0.4 v			4A		-1.6	mA
			36												5.5 v	0.4 v		4B		-1.6	mA

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TABLE III. Group A inspection for device type 07. - Continued

Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 2 / 1A	2 / 3 / 1B	3 / 1 / 1Y	4 / 14 / V_{CC}	5 / 4 / 2Y	6 / 5 / 2A	7 / 6 / 2B	8 / 10 / 3Y	9 / 8 / 3A	10 / 9 / 3B	11 / 7 / GND	12 / 11 / 4A	13 / 12 / 4B	14 / 13 / 4Y	Measured terminal	Test limits Min	Test limits Max	Unit
1																					
T_C = 25°C	I_{CCL}	3005	37	5.5 v	5.5 v		5.5 v		5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		V_{CC}		20	mA
	I_{CCH}	3005	38	GND	GND		5.5 v		GND	GND		GND	GND		GND	GND		V_{CC}		6.6	mA
	V_{IC}		39	-12 mA			4.5 v											1A		-1.5	V
			40		-12 mA													1B		-1.5	V
			41						-12 mA									2A		-1.5	V
			42							-12 mA								2B		-1.5	V
			43									-12 mA						3A		-1.5	V
			44										-12 mA					3B		-1.5	V
			45												-12 mA			4A		-1.5	V
			46													-12 mA		4B		-1.5	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003	47	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y	3	23	ns
T_C = 25°C		(Fig. 4)	48															2A to 2Y			ns
			49															3A to 3Y			ns
			50															4A to 4Y			ns
	t_{PLH}		51	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y		28	ns
			52															2A to 2Y			ns
			53															3A to 3Y			ns
			54															4A to 4Y			ns
10	t_{PHL}	3003	55	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y		29	ns
T_C = 125°C		(Fig. 4)	56															2A to 2Y			ns
			57															3A to 3Y			ns
			58															4A to 4Y			ns
	t_{PLH}		59	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y		35	ns
			60															2A to 2Y			ns
			61															3A to 3Y			ns
			62															4A to 4Y			ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 08.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,D / Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	2.0 v		5.5 v	4.5 v	5.5 v		5.5 v		5.5 v				5.5 v	16 mA	1Y		0.4	V
			2	5.5 v	16 mA	2.0 v		5.5 v										2Y		0.4	V
			3			5.5 v		2.0 v	16 mA	2.0 v	16 mA	2.0 v	16 mA					3Y		0.4	V
			4					5.5 v		5.5 v		5.5 v						4Y		0.4	V
			5															5Y		0.4	V
			6												16 mA	2.0 v		6Y		0.4	V
	I_{CEX}		7	0.8 v												5.5 v	5.5 v	1Y		250	µA
			8	5.5 v	5.5 v	0.8 v		0.8 v	5.5 v	0.8 v	5.5 v	0.8 v	5.5 v					2Y		250	µA
			9			5.5 v		5.5 v		5.5 v		5.5 v				0.8 v		3Y		250	µA
			10															4Y		250	µA
			11															5Y		250	µA
			12												5.5 v			6Y		250	µA
	V_{IC}		13	-12 mA														1A		-1.5	V
			14			-12 mA		-12 mA		-12 mA		-12 mA						2A		-1.5	V
			15															3A		-1.5	V
			16															4A		-1.5	V
			17															5A		-1.5	V
			18													-12 mA		6A		-1.5	V
	I_{IH1}	3010	19	2.4 v		GND	5.5 v	GND		GND		GND				GND		1A		40	µA
			20	GND		2.4 v		GND										2A		40	µA
			21			GND		2.4 v		2.4 v		2.4 v				2.4 v		3A		40	µA
			22					GND		GND		GND						4A		40	µA
			23															5A		40	µA
			24															6A		40	µA
	I_{IH2}		25	5.5 v		5.5 v		5.5 v		5.5 v		5.5 v				GND		1A		100	µA
			26	GND		GND		GND										2A		100	µA
			27															3A		100	µA
			28															4A		100	µA
			29															5A		100	µA
			30															6A		100	µA
	I_{IL}	3009	31	0.4 v		5.5 v		5.5 v		5.5 v		5.5 v				5.5 v		1A	-0.7	-1.6	mA
			32	5.5 v		0.4 v		5.5 v										2A		-1.6	mA
			33			5.5 v		0.4 v		0.4 v		0.4 v				0.4 v		3A		-1.6	mA
			34					5.5 v		5.5 v		5.5 v						4A		-1.6	mA
			35															5A		-1.6	mA
			36															6A		-1.6	mA

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TABLE III. Group A inspection for device type 08. - Continued

Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Case C	1	4	3	14	5	6	9	8	11	10	7	12	13	2				
			Test No.	1A	2Y	2A	V_{CC}	3A	3Y	4A	4Y	5A	5Y	GND	6Y	6A	1Y				
1 T_C 25°C	I_{CCL}	3005	37	5.5 v		5.5 v	5.5 v	5.5 v		5.5 v		5.5 v				5.5 v		V_{CC}		30	mA
	I_{CCH}	3005	38	GND		GND	5.5 v	GND		GND		GND				GND		V_{CC}		9.9	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125° C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55° C and V_{IC} tests are omitted.																				
9 T_C 25°C	t_{PHL}	3003 (Fig. 4)	39	IN	OUT	IN	5.0 v	IN	OUT	IN	OUT	IN	OUT	GND	OUT	IN	OUT	1A to 1Y	3	23	ns
			40															2A to 2Y			ns
			41															3A to 3Y			ns
			42															4A to 4Y			ns
			43															5A to 5Y			ns
			44															6A to 6Y			ns
	t_{PLH}		45	IN	OUT	IN		IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	1A to 1Y		28	ns
			46															2A to 2Y			ns
			47															3A to 3Y			ns
			48															4A to 4Y			ns
			49															5A to 5Y			ns
			50															6A to 6Y			ns
10 T_C = 125°C	t_{PHL}		51	IN	OUT	IN		IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	1A to 1Y		29	ns
			52															2A to 2Y			ns
			53															3A to 3Y			ns
			54															4A to 4Y			ns
			55															5A to 5Y			ns
			56															6A to 6Y			ns
	t_{PLH}		57	IN	OUT	IN		IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	1A to 1Y		35	ns
			58															2A to 2Y			ns
			59															3A to 3Y			ns
			60															4A to 4Y			ns
			61															5A to 5Y			ns
			62															6A to 6Y			ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 09.

Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case C / Test No.	1 / 1A	2 / 1B	3 / 1Y	4 / 2A	5 / 2B	6 / 2Y	7 / GND	8 / 3Y	9 / 3A	10 / 3B	11 / 4Y	12 / 4A	13 / 4B	14 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	2.0 v	2.0 v	16 mA	5.5 v	5.5 v		GND		5.5 v	5.5 v		5.5 v	5.5 v	4.5 v	1Y		0.4	v
			2	5.5 v	5.5 v		2.0 v	2.0 v	16 mA			5.5 v	5.5 v					2Y			
			3				5.5 v	5.5 v			16 mA	2.0 v	2.0 v					3Y			
			4									5.5 v	5.5 v	16 mA	2.0 v	2.0 v		4Y			
	I_{CEX}		5	0.8 v	4.5 v	5.5 v									5.5 v	5.5 v		1Y		250	A
			6	4.5 v	0.8 v	5.5 v												1Y			
			7	5.5 v	5.5 v		0.8 v	4.5 v	5.5 v									2Y			
			8				4.5 v	0.8 v	5.5 v									2Y			
			9				5.5 v	5.5 v			5.5 v	0.8 v	4.5 v					3Y			
			10								5.5 v	4.5 v	0.8 v					3Y			
			11									5.5 v	5.5 v	5.5 v	0.8 v	4.5 v		4Y			
			12									5.5 v	5.5 v	5.5 v	4.5 v	0.8 v		4Y			
	V_{IC}		13	-12 mA														1A		-1.5	V
			14		-12 mA													1B			
			15				-12mA											2A			
			16					-12 mA										2B			
			17									-12mA						3A			
			18										-12 mA					3B			
			19												-12 mA			4A			
			20													-12mA		4B			
	I_{IH_1}	3010	21	2.4 v	GND		GND	GND				GND	GND		GND	GND	5.5 v	1A		40	A
			22	GND	2.4 v		GND											1B			
			23		GND		2.4 v											2A			
			24				GND	2.4 v										2B			
			25					GND				2.4 v	2.4 v					3A			
			26									GND	GND					3B			
			27												2.4 v			4A			
			28												GND	2.4 v		4B			
	I_{IH_2}	3010	29	5.5 v												GND		1A		100	A
			30	GND	5.5 v													1B			
			31		GND		5.5 v											2A			
			32				GND	5.5 v										2B			
			33					GND				5.5 v						3A			
			34									GND	5.5 v					3B			
			35										GND		5.5 v			4A			
			36										GND		GND	5.5 v		4B			

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TABLE III. Group A inspection for device type 09. -Continued

Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits		Unit
			Test No.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}		Min	Max	
1 T_C = 25°C	I_{IL}	3009	37	0.4 v	5.5 v		5.5 v	5.5 v		GND		5.5 v	5.5 v		5.5 v	5.5 v	5.5 v	1A	-0.7	-1.6	mA
			38	5.5 v	0.4 v		5.5 v											1B			
			39		5.5 v		0.4 v											2A			
			40				5.5 v	0.4 v										2B			
			41					5.5 v				0.4 v						3A			
			42									5.5 v	0.4 v					3B			
			43										5.5 v		0.4 v			4A			
			44												5.5 v	0.4 v		4B			
	I_{CCL}	3005	45												5.5 v	5.5 v				20	mA
	I_{CCH}	3005	46	GND	GND		GND	GND				GND	GND		GND	GND				6.6	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125° C, and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55° C, and V_{IC} tests are omitted.																				
9	t_{PHL}	3003 (Fig. 4)	47	IN	2.4 v	OUT	IN	2.4 v	OUT	GND	OUT	IN	2.4 v	OUT	IN	2.4 v	5.0 v	1A to 1Y	3	23	ns
			48															2A to 2Y			
			49															3A to 3Y			
			50															4A to 4Y			
	t_{PLH}		51	IN	2.4 v	OUT	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		1A to 1Y		28	
			52															2A to 2Y			
			53															3A to 3Y			
			54															4A to 4Y			
10 T_C = 125°C	t_{PHL}		55	IN	2.4 v	OUT	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		1A to 1Y		29	
			56															2A to 2Y			
			57															3A to 3Y			
			58															4A to 4Y			
	t_{PLH}		59	IN	2.4 v	OUT	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		1A to 1Y		35	
			60															2A to 2Y			
			61															3A to 3Y			
			62															4A to 4Y			
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
V_{IN}	Voltage level at an input terminal
V_{IC}	Input clamp voltage
I_{IN}	Current flowing into an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	5430
02	5420
03	5410
04	5400
05	5404
06	5412
07	5401
08	5405
09	5403

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - CG, MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0496)

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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